

Pima Indians Diabetes Analysis

Internship Project Report — Task 1

1. Introduction

Diabetes is a chronic condition affecting millions of people worldwide, and early detection can significantly improve patient outcomes through timely intervention. This project builds a machine learning pipeline that predicts whether a patient is likely to have diabetes based on routinely available diagnostic measurements, using the well-known Pima Indians Diabetes Dataset.

2. Objective

The objective of this project is to perform exploratory data analysis on the dataset, handle missing/implausible values, train and compare multiple classification models, and select the best-performing model for deployment as a reusable prediction module.

3. Dataset Description

The dataset contains 768 records of female patients of Pima Indian heritage, aged 21 and older, with 8 diagnostic input features and a binary target (Outcome: 1 = diabetic, 0 = not diabetic).

Feature	Description
Pregnancies	Number of times pregnant
Glucose	Plasma glucose concentration (2-hr OGTT)
BloodPressure	Diastolic blood pressure (mm Hg)
SkinThickness	Triceps skin fold thickness (mm)
Insulin	2-Hour serum insulin (mu U/ml)
BMI	Body mass index (kg/m ²)
DiabetesPedigreeFunction	Diabetes likelihood from family history
Age	Age in years
Outcome	Target label (1 = diabetic, 0 = not diabetic)

4. Data Preprocessing

Several clinical columns (Glucose, BloodPressure, SkinThickness, Insulin, BMI) encode missing measurements as 0, which is not physiologically valid. These zero values were replaced with the median of their respective column. The dataset was then split into training (80%) and testing (20%) sets using stratified sampling to preserve class balance, and all features were standardized using scikit-learn's

StandardScaler.

5. Methodology

Three classification algorithms were trained and evaluated on the same preprocessed data: Logistic Regression, Support Vector Machine (SVM) with an RBF kernel, and Random Forest (200 trees, max depth 6). Each model's performance was measured using accuracy on the held-out test set, and the best-performing model was selected automatically.

6. Results

Model	Test Accuracy
Logistic Regression	70.78%
Support Vector Machine (Best)	74.03%
Random Forest	73.38%

The Support Vector Machine achieved the best test accuracy of 74.03%, with an ROC-AUC of approximately 0.80, indicating good discriminative ability between diabetic and non-diabetic patients.

7. Visualizations

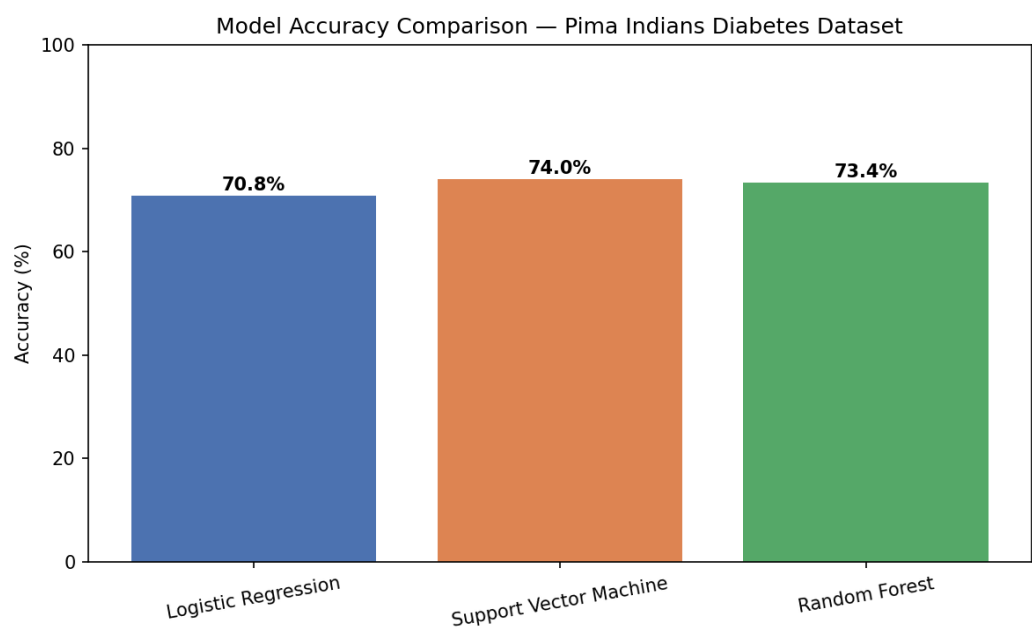


Figure 1: Model accuracy comparison across the three classifiers.

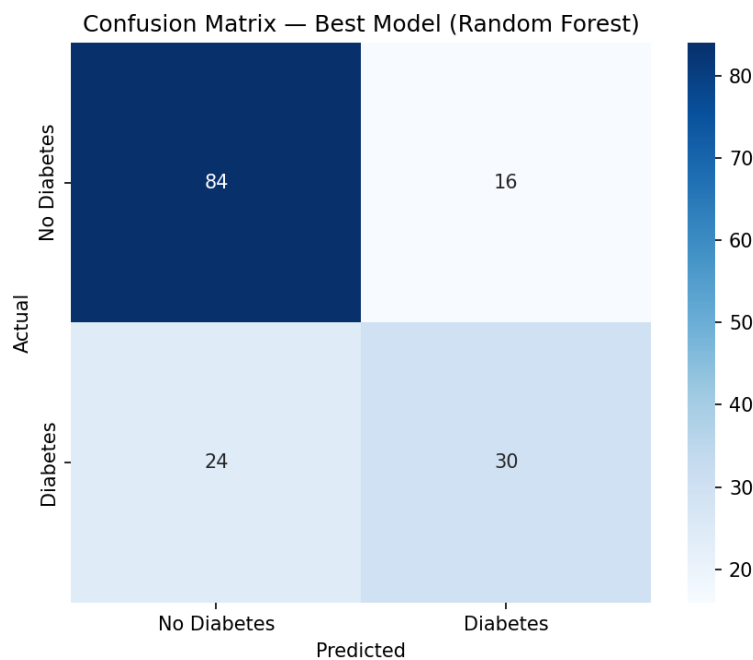


Figure 2: Confusion matrix of the best model (SVM) on the test set.

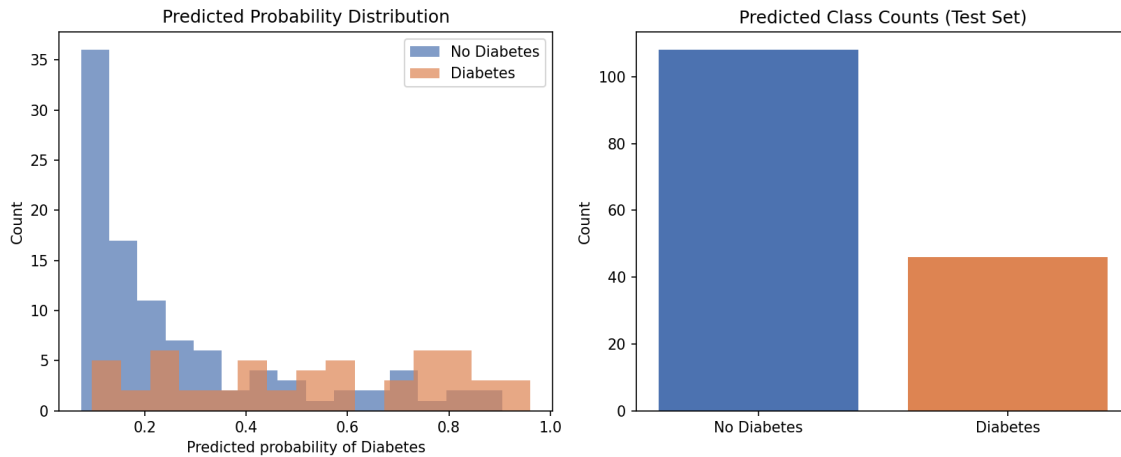


Figure 3: Predicted probability distribution by true class, and predicted class counts on the test set.

8. Conclusion

This project demonstrates a complete, reproducible machine learning workflow: from raw data cleaning through model comparison to a saved, deployable model artifact. The Support Vector Machine model, combined with median imputation and feature scaling, provided the best balance of accuracy and generalization on this dataset. The trained model, along with its fitted scaler, is packaged in `outputs/trained_model.pkl` and can be used directly via `src/prediction.py` to make predictions on new patient data.

9. Future Improvements

Potential enhancements include hyperparameter tuning via grid/randomized search, handling class imbalance with techniques such as SMOTE, trying gradient boosting models (e.g. XGBoost/LightGBM), and deploying the model behind a simple web API for real-time predictions.